

- 3 (a) (i) amplitude = 0.5 cm A1 [1]
- (ii) period = 0.8 s A1 [1]
- (b) (i) $\omega = 2\pi / T$ C1
 $= 7.85 \text{ rad s}^{-1}$
 correct use of $v = \omega \sqrt{(x_0^2 - x^2)}$ B1
 $= 7.85 \times \sqrt{\{0.5 \times 10^{-2}\}^2 - \{0.2 \times 10^{-2}\}^2}$
 $= 3.6 \text{ cm s}^{-1}$ A1 [3]
(if tangent drawn or clearly implied (B1)
 $3.6 \pm 0.3 \text{ cm s}^{-1}$ (A2)
but allow 1 mark for $> \pm 0.3$ but $\leq \pm 0.6 \text{ cm s}^{-1}$)
- (ii) $d = 15.8 \text{ cm}$ A1 [1]
- (c) (i) (continuous) loss of energy / reduction in
 amplitude (from the oscillating system) B1
 caused by force acting in opposite direction to the motion / friction /
 viscous forces B1 [2]
- (ii) same period / small increase in period B1
 line displacement always less than that on Fig.3.2 (*ignore first $T/4$*) M1
 peak progressively smaller A1 [3]